

MLOps Foundations Exercises

SRAI Book 5 Lesson 1 | PU-B05-C01 | Exercises v0.1

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These exercises convert the lesson into applied evidence. Use the validated companion notebook and synthetic data. Do not use the outputs for real decisions about farms, learners, patients, households or communities.

Submission requirements

- Preserve seed 42 for canonical reproduction.
- Show intermediate calculations and report metrics to three decimals.
- Name every evidence artifact and accountable owner.
- Keep governance approval false during the first canonical run.
- Distinguish observed results, interpretation, recommendation and authorization.

Exercise 1 Governed lifecycle

Map the lifecycle for a monthly public-sector prediction service. Identify states, artifacts, owners, mandatory gates, exception authority and return paths. Apply it to one of Agriculture, Education, Health or Habitat.

Required evidence

- Written reasoning tied to the declared use and population.
- A versioned artifact or notebook output that supports the claim.
- At least one failure condition and its prescribed response.
- A short limitation statement identifying what the evidence does not establish.

Cross-sector transfer question

Explain one material change required if the same technical component moved to another SRAI sector. Address data validity, consequence, authority or monitoring rather than merely renaming fields.

Exercise 2 Reproducibility contract

Write a contract covering data, code, configuration, environment, random state, hardware assumptions and tolerance. Explain what a cryptographic digest proves and what it cannot prove.

Required evidence

- Written reasoning tied to the declared use and population.
- A versioned artifact or notebook output that supports the claim.
- At least one failure condition and its prescribed response.
- A short limitation statement identifying what the evidence does not establish.

Cross-sector transfer question

Explain one material change required if the same technical component moved to another SRAI sector. Address data validity, consequence, authority or monitoring rather than merely renaming fields.

Exercise 3 Data acceptance

Design schema, range, missingness, temporal and category checks for the selected sector. Inject one failure in the notebook and show that training stops.

Required evidence

- Written reasoning tied to the declared use and population.
- A versioned artifact or notebook output that supports the claim.
- At least one failure condition and its prescribed response.
- A short limitation statement identifying what the evidence does not establish.

Cross-sector transfer question

Explain one material change required if the same technical component moved to another SRAI sector. Address data validity, consequence, authority or monitoring rather than merely renaming fields.

Exercise 4 Evaluation decision

Using a confusion matrix with TP 160, FP 80, FN 40 and TN 220, calculate precision, recall, specificity, accuracy and F1. Recommend a threshold policy linked to consequences and intervention capacity.

Required evidence

- Written reasoning tied to the declared use and population.
- A versioned artifact or notebook output that supports the claim.
- At least one failure condition and its prescribed response.
- A short limitation statement identifying what the evidence does not establish.

Cross-sector transfer question

Explain one material change required if the same technical component moved to another SRAI sector. Address data validity, consequence, authority or monitoring rather than merely renaming fields.

Exercise 5 Lineage and registry

Create a manifest that binds dataset, split, code, environment, model, evaluation and decision records. Distinguish registration, eligibility for review, approval and deployment.

Required evidence

- Written reasoning tied to the declared use and population.
- A versioned artifact or notebook output that supports the claim.
- At least one failure condition and its prescribed response.
- A short limitation statement identifying what the evidence does not establish.

Cross-sector transfer question

Explain one material change required if the same technical component moved to another SRAI sector. Address data validity, consequence, authority or monitoring rather than merely renaming fields.

Exercise 6 Service and monitoring

Specify an inference contract and a monitoring plan covering service, data, prediction, outcome, subgroup, security and cost indicators. Assign owners and responses.

Required evidence

- Written reasoning tied to the declared use and population.
- A versioned artifact or notebook output that supports the claim.
- At least one failure condition and its prescribed response.
- A short limitation statement identifying what the evidence does not establish.

Cross-sector transfer question

Explain one material change required if the same technical component moved to another SRAI sector. Address data validity, consequence, authority or monitoring rather than merely renaming fields.

Exercise 7 Incident and rollback

Respond to a silent unit change in a mandatory input. Define detection, containment, impact assessment, rollback, recovery verification and post-incident evidence.

Required evidence

- Written reasoning tied to the declared use and population.
- A versioned artifact or notebook output that supports the claim.
- At least one failure condition and its prescribed response.
- A short limitation statement identifying what the evidence does not establish.

Cross-sector transfer question

Explain one material change required if the same technical component moved to another SRAI sector. Address data validity, consequence, authority or monitoring rather than merely renaming fields.

Exercise 8 Governance and retirement

Explain why a technically ready model remains blocked without approval. Define authorized remediation, review expiry and retirement evidence.

Required evidence

- Written reasoning tied to the declared use and population.
- A versioned artifact or notebook output that supports the claim.
- At least one failure condition and its prescribed response.
- A short limitation statement identifying what the evidence does not establish.

Cross-sector transfer question

Explain one material change required if the same technical component moved to another SRAI sector. Address data validity, consequence, authority or monitoring rather than merely renaming fields.

Assessment summary

Exercise	Marks
Exercise 1 Governed lifecycle	12
Exercise 2 Reproducibility contract	12
Exercise 3 Data acceptance	12
Exercise 4 Evaluation decision	16
Exercise 5 Lineage and registry	12
Exercise 6 Service and monitoring	14
Exercise 7 Incident and rollback	10
Exercise 8 Governance and retirement	12
Total	100